# CRUSOE x POLAR INFRASTRUCTURE

*Joint Venture: Nordic AI Factory Expansion | Site: OSL-HAMAR-01*

# PRE-CONSTRUCTION RESOURCE & ENERGY ESTIMATES

## 1. PROJECT IDENTIFICATION & SCOPE

Site ID: OSL-HAMAR-01 (Mainland Europe AI Hub)
Partners: Crusoe Energy Systems & Polar Infrastructure
Design Capacity: 12.0 Megawatts (MW) IT Load (Expandable to 52MW)
Campus Footprint: Approx. 130,000 sq ft (Facility Hall A)
Cooling Architecture: Direct-to-Chip Liquid Cooling (DLC) mapped for NVIDIA Blackwell architecture.
Target Date of Energization: Q4 2026

## 2. PHASE 1: PLANNED MATERIAL PROCUREMENT (EMBODIED)

The following estimates outline the physical materials required to construct the facility. Procurements favor regional Nordic suppliers.

| Material / Component Specification | Estimated Quantity (Planned) |
|---|---|
| Structural Steel (Nordic Green Steel) | 3,200 metric tons |
| Low-Carbon Concrete (Foundation Mix) | 9,500 cubic yards |
| Copper Cabling & Busbars | 240 metric tons |
| DLC Polymer Piping (PVC-free) | 22,000 linear meters |
| NVIDIA GB200 Blackwell Racks | 42 x NVL72 Racks (3,024 GPUs) |
| Mechanical Skids & Exchangers | 12 x 1MW Skids |

## 3. PHASE 2: OPERATIONAL RESOURCE FORECAST (YEAR 1)

Estimates assume a 100% load factor for 24/7 high-performance AI training workloads (12MW IT load + cooling overhead). The data below represents raw energy and resource demands for external auditing and carbon calculation.

| Operational Resource Vector | Projected Annual Consumption |
|---|---|
| Facility Electricity (Grid Draw) | 111.4 GWh / year |
| Primary Electrical Fuel Source | 100% Norwegian Regional Hydroelectric Grid |
| Generator Testing Fuel (HVO100 Renewable Diesel) | 15,000 Liters / year |
| Facility Cooling Water (Closed Loop DLC) | 5,250,000 Liters / year |
| Surplus Thermal Energy (Export to District Heating) | 88.0 GWh / year (Recovered Heat) |

Target Facility PUE (Power Usage Effectiveness): 1.06

## 4. NORWEGIAN REGULATORY & COMPLIANCE STATEMENT

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This facility has been designed to strictly adhere to the updated Norwegian regulatory landscape of 2025/2026, ensuring immediate compliance and mitigating legal friction for our VC partners and AI tenants:

- Data Centre Regulation (Jan 1, 2025): The site is fully registered with the Norwegian Communications Authority (Nkom). We have implemented the mandated Security Management Systems, ensuring physical and digital resilience against geopolitical threats, including strict KYC protocols for all AI tenants.

- Surplus Heat Recovery (April 1, 2025 Energy Act): Complying with the new mandate for facilities over 2MW, a full Cost-Benefit Analysis (CBA) has been executed. We have secured a symbiotic offtake agreement to export 88 GWh of surplus thermal energy to the Hamar district heating network and local agricultural drying facilities.

- European Directives: The facility's transparent lifecycle reporting ensures our tenants easily comply with the EU AI Act (Article 40) environmental disclosures and the Corporate Sustainability Reporting Directive (CSRD) Scope 3 requirements.